

Maya V. Chung

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[Professional Website](#)

[Google Scholar](#)

[Github](#)

EDUCATION	Ph.D. Atmospheric and Oceanic Sciences, Princeton University (exp.) 2025 Graduate certificate in Science, Technology and Environmental Policy (in progress) Teaching transcript (in progress) B.A. Earth and Planetary Sciences, Harvard University 2015–2019 <i>magna cum laude with highest honors</i> Secondary Field: Mathematical Sciences Language Citation: Mandarin Chinese
RESEARCH EXPERIENCE	Graduate Research Assistant 2020–Present <i>Princeton University Program in Atmospheric and Oceanic Sciences</i> Advised by Gabriel Vecchi Committee: Laure Resplandy, Andrew Wittenberg, Rong Zhang, Sonya Legg (former) • Researching interactions between ocean salinity variability, the tropical Pacific mean state and the El Niño–Southern Oscillation (ENSO) using coupled general circulation models (GCMs) and observations. • Investigating the climate system’s response to warming and cooling in coupled GCMs, including ocean circulation changes and climate feedbacks. • Exploring extreme heat and cold-related mortality across different climate regimes HMEI-STEP Fellow 2022–2024 <i>High Meadows Environmental Institute, Princeton School of Public and International Affairs</i> Advised by Jessica Metcalf and Bryan Grenfell Integrated climate and infectious disease modeling to study year-to-year variations in disease spread associated with ENSO. Undergraduate Research Assistant 2017–2019 <i>Harvard Department of Earth and Planetary Sciences</i> Advised by Peter Huybers Developed a novel warping technique to identify ocean profiles influenced by water mass intrusions and improve estimates of global ocean heat content. Projected variability onto seasonal modes and the El Niño–Southern Oscillation to discern heating trends and interannual changes in ocean heat content. Summer Student Fellow Summer 2018 <i>Woods Hole Oceanographic Institution</i> Advised by Geoffrey Gebbie and Peter Huybers Created a novel method to measure ocean heat uptake by adapting an algorithm commonly used for speech recognition (dynamic time warping). Summer Undergraduate Research Fellow Summer 2017 <i>Scripps Institution of Oceanography</i> Advised by Ivana Cerovecki, Matthew Mazloff, Sarah Gille, Lynne Talley Characterized relationships between sea ice production and winds in West Antarctica and their connection to cooling and freshening events in the southeast Pacific. Compared model output to satellite and in-situ observations.

UNIVERSITY TEACHING & LEADERSHIP	<i>ENV 354/GEO 368: Climate and Weather: Order in the Chaos</i> Fall 2025 Guest Lecturer, Princeton University Professor: Gabriel Vecchi Guest lecture on the El Niño–Southern Oscillation for undergraduates. <i>SPI 586D: Global Environmental Governance</i> Spring 2024, 2025 Guest Lecturer, Princeton University and NYU School of Law Professors: Michael Oppenheimer, Bryce Rudyk Guest lecture on solar radiation management, climate modeling, and international governance strategies for geoengineering. Professional Development Associate, GradFUTURES, Princeton University 2024-25 Supported science policy career pathways and professional development opportunities for Princeton graduate students. Co-organized the inaugural science policy learning cohort, science communication workshop, and evaluated AAAS CASE workshop applications. Advisor, High Meadows Environmental Institute Summer Internship, Princeton University 2024-25 Advised undergraduate researcher Emma Dornseif during the summer and senior thesis project on extreme temperature and mortality, joint with the Department of Anthropology.
	<i>GEO 425 / MAE 425: Introduction to Ocean Physics for Climate</i> Fall 2022 Assistant in Instruction & Guest Lecturer, Princeton University Professor: Gabriel Vecchi Guest lectures (5): Surface heat fluxes / ocean heat budget, mixed layer dynamics, deep ocean circulation and global overturning climate impacts, ENSO observations, ENSO theory.
	<i>USW 35: Dilemmas of Equity and Excellence in American K-12 Education</i> Fall 2018 Course Assistant, Harvard College Program in General Education Professor: Katherine (Kay) Merseth Topics: US public education history, current issues, and reform.
	<i>APMTH 120: Applied Linear Algebra and Big Data</i> Spring 2018 Teaching Fellow, Harvard School of Engineering & Applied Sciences Professor: Eli Tziperman Topics: singular value decomposition, spectral clustering, neural networks, applications to economics & science.
	AmeriCorps Member 2019–2020 <i>City Year Boston, Henry Grew Elementary School</i> Provided holistic support and tracked student data in a 5th grade classroom, including individual and small-group tutoring in math and English language arts, social-emotional coaching, after-school enrichment, and school-wide programming.
OTHER TEACHING EXPERIENCE	Freelance Tutor 2019–2020 Tutored high school students, undergraduates, and adult learners. Subjects: Earth Science, Chinese, MATLAB, college essay writing, SAT prep.
	Science Teacher Summer 2019 <i>Children’s School of Science, Woods Hole, MA</i> Designed curricula and taught hands-on, field-based summer courses for students ages 8-12. Courses: Marine Biology, Invertebrate Zoology.
	Peer Tutor 2016–2019 <i>Harvard College Bureau of Study Counsel</i> Tutored 11 undergraduates in linear algebra, multivariable calculus, statistics, probability, Chinese, and American Sign Language.

AWARDS & HONORS	First Place, Health in Climate Hackathon , Climate Week NYC, Cornell Tech	2025
	Created an interactive AI agent that sends actionable weather and health alerts via text message before, during, and after extreme weather events.	
	Outstanding Student Presentation Award , American Geophysical Union	2025
	Service and Outreach Award , Princeton Geosciences Department	2022
	magna cum laude with highest honors	2019
	Highest honors for research within the field of Earth and Planetary Sciences at Harvard College.	
	Thomas T. Hoopes Prize	2019
FELLOWSHIPS	Awarded to Harvard seniors nominated by faculty for conducting outstanding senior thesis research.	
	Student Travel Grant , American Geophysical Union Fall Meeting	2018
	Student Travel Grant , Ocean Sciences Meeting	2018
	High Meadows Environmental Institute Science, Technology, and Environmental Policy Graduate Fellowship (HMEI-STEP)	2022–2024
	National Science Foundation Graduate Research Fellowship (NSF GRFP)	2021–2024
	American Meteorological Society Graduate Fellowship	2020–2021
	Woods Hole Oceanographic Institution Summer Student Fellowship	2018
PEER-REVIEWED PUBLICATIONS	Scripps Institution of Oceanography SURF REU	2017
	<i>(In revision)</i> Chung, M. V. , Liu, M., Soden, B. J., Vecchi, G. A. The influence of sea surface salinity variability on the equatorial Pacific mean state and extreme El Niño events.	
	3. <i>(Accepted)</i> Chung, M. V. , Yang, W., Vecchi, G. A. (2025). Runaway Cooling from Large Solar Reductions Modulated by Ocean Overturning Circulation and Heat Uptake. <i>Geophysical Research Letters</i> . Preprint: https://essopenarchive.org/doi/full/10.22541/essoar.175190999.97949534/v2	
	2. Chung, M. V. , Vecchi, G. A., Yang, W., Grenfell, B., and Metcalf, C. J. (2025). Intersecting memories of immunity and climate: Potential multiyear impacts of the El Niño-Southern Oscillation on infectious disease spread. <i>GeoHealth</i> , 9, e2024GH001193. https://doi.org/10.1029/2024GH001193	
	1. Knutson, T. R., Chung, M. V. , Vecchi, G., Sun, J., Hsieh, T-L. and Smith, A. J. P. (2021). ScienceBrief Review: Climate change is probably increasing the intensity of tropical cyclones. <i>Critical Issues in Climate Change Science</i> . https://doi.org/10.5281/zenodo.4570334	
	2. <i>(Book Chapter)</i> Yang, W., Levin, E., Menemenlis, S., Scapin, N., Igbino, M., Chung, M. , Rios, G., Hsieh, T.-L., Deike, L., Mitevski, I., & Vecchi, G. A. (2025). Chapter 1—Overview of tropical cyclones and historical perspective. In G. Villarini, G. A. Vecchi, & E. Scoccimarro (Eds.), <i>Tropical Cyclones and Associated Impacts</i> (pp. 1–25). Elsevier. https://doi.org/10.1016/B978-0-323-95390-0.00001-7	
	1. <i>(Senior thesis, award)</i> Chung, M. V. , Gebbie, G., and Huybers, P. J. (2019). Quantifying Isopycnal Heave Using Dynamic Depth Warping (Senior thesis, Harvard College, Cambridge, MA). https://eps.harvard.edu/files/eps/files/mayachung_thesis_final.pdf	
OTHER PUBLICATIONS		

CONFERENCE TALKS	(invited) M. V. Chung, 2024: “Infectious Disease Dynamics in a Changing Climate.” <i>Detection and Attribution Science Workshop</i>, University of the West Indies, Mona, Kingston, Jamaica. M. V. Chung, W. Yang, and G. A. Vecchi, 2023: “Abrupt Solar Changes Yield Asymmetric Responses in Deep Water Density and Meridional Overturning Circulation.” <i>American Geophysical Union Fall Meeting</i>, San Francisco, CA. M. V. Chung, W. Yang, G. A. Vecchi, B. Grenfell, and C. J. Metcalf, 2023: “Multiyear Impacts of ENSO on Infectious Disease Spread in Weather-Disease Models.” <i>American Geophysical Union Fall Meeting</i>, San Francisco, CA. M. V. Chung, M. Liu, B. Soden, and G. A. Vecchi, 2023: “The influence of sea surface salinity variability on the equatorial Pacific mean state and extreme ENSO events.” <i>Northeast Tropical Workshop</i>, University at Albany, Albany, NY. M. V. Chung, M. Liu, B. Soden, and G. A. Vecchi, 2022: “The Role of Sea Surface Salinity in Extreme El Niño events.” <i>Ocean Salinity Conference</i>, Columbia University, New York, NY. M. V. Chung, M. Liu, B. Soden, and G. A. Vecchi, 2021: “The Role of Sea Surface Salinity in Extreme El Niño events.” <i>American Geophysical Union Fall Meeting</i>, New Orleans, LA.
OTHER FORMAL TALKS	(invited) M. V. Chung, W. Yang, G. A. Vecchi, 2025: “Runaway Cooling from Large Solar Reductions Modulated by Ocean Overturning Circulation and Heat Uptake.” <i>45th ECS & Cloud Feedback Virtual Symposium</i>. https://youtu.be/YJK-MvE-Pys?si=_X09i8-RqB4484r9&t=550 (invited) M. V. Chung, 2024: “ENSO forecasting and implications for infectious disease prediction.” <i>On the accuracy (and niceness) of prediction: From epidemics to climate and weather</i>, High Meadows Environmental Institute and Department of Ecology and Evolutionary Biology, Princeton, NJ. M. V. Chung, W. Yang, G. A. Vecchi, B. Grenfell, and C. J. Metcalf, 2023: “Multiyear Impacts of ENSO on Infectious Disease Spread in Weather-Disease Models.” <i>STEP Seminar</i>, School for Public and International Affairs, Princeton, NJ. M. V. Chung, 2023: “Climate Science at the IPCC: How Research Informs Policy.” <i>John Locke Institute Public Policy Symposium</i>, Princeton, NJ. M. V. Chung, 2019: “Down with Density: A New Way to Quantify Ocean Warming due to Climate Change.” <i>Harvard College 3-Minute Thesis Finalist Presentations</i>, Harvard College Writing Center, Cambridge, MA. M. V. Chung, G. Gebbie, and P. Huybers, 2019: “Quantifying Isopycnal Heave Using Dynamic Depth Warping.” <i>2019 Senior Thesis Presentations</i>, Harvard University Department of Earth and Planetary Sciences, Cambridge, MA. M. V. Chung, G. Gebbie, and P. Huybers, 2018: “Quantifying Layer Thickness Changes Using Dynamic Depth Warping.” <i>Summer Student Fellow Presentations</i>, Woods Hole Oceanographic Institution Physical Oceanography Department, Woods Hole, MA.
POSTER PRESENTATIONS	(Award) M. V. Chung, W. Yang, G. A. Vecchi, B. Grenfell, and C. J. Metcalf, 2024: “Intersecting memories of immunity and climate: Potential multiyear impacts of the El Niño—Southern Oscillation on infectious disease spread.” <i>American Geophysical Union Fall Meeting</i>, Washington, D.C. M. V. Chung, W. Yang, G. A. Vecchi, B. Grenfell, and C. J. Metcalf, 2024: “Intersecting memories of immunity and climate: Potential multiyear impacts of the El Niño—Southern Oscillation on infectious disease spread.” <i>HMEI Discovery Day</i>, Princeton, NJ. (invited) M. V. Chung, M. Liu, B. Soden, and G. A. Vecchi, 2023: “The Influence of Sea Surface Salinity on the Equatorial Pacific Mean State and Extreme ENSO Events.” <i>American Geophysical Union Fall Meeting</i>, San Francisco, CA. M. V. Chung, M. Liu, B. Soden, and G. A. Vecchi, 2022: “The Influence of Sea Surface Salinity on the Equatorial Pacific Mean State and Extreme ENSO Events.” <i>American Geophysical Union Fall Meeting</i>, Chicago, IL.

M. V. Chung, G. Gebbie, and P. Huybers, 2020: “Quantifying Ocean Heat Content Changes Related to ENSO, Seasonal Variability, and Trends in Isopycnal Heave.” *Ocean Sciences Meeting*, San Diego, CA.

M. V. Chung, G. Gebbie, and P. Huybers, 2018: “Quantifying Isopycnal Heave Using Dynamic Depth Warping.” *American Geophysical Union Fall Meeting*, Washington, D. C.

M. V. Chung, I. Cerovecki, F. A. Haumann, M. Mazloff, S. Gille, and L. Talley, 2018: “Variability of Sea Ice Production in the Ross Sea from 2006-2010 and its Relationship to the Amundsen Sea Low.” *Ocean Sciences Meeting*, Portland, OR.

M. V. Chung, I. Cerovecki, M. Mazloff, S. Gille, and L. Talley, 2018: “Variability of Ice Production in the Ross Sea in 2006-2010 and its Relationship to the Amundsen Sea Low.” *SURF Research Symposium*, Scripps Institution of Oceanography, La Jolla, CA.

MEDIA ["The Interplay of ENSO and Immunity in Infectious Disease Outbreaks"](#) (research highlight) - *Eos*, March 2025

PhD Candidate Maya Chung in conversation with Michael Igbinoba (interview) - *Princeton's Climate Stories: A My Climate Story Magazine*, Issue 1, Spring 2025

["Why Hurricanes And Typhoons Will Become More Dangerous"](#) (quoted) – *Forbes*, March 2021

["Intensity of Tropical Cyclones is Probably Increasing Due to Climate Change"](#) (quoted) – University of East Anglia, March 2021

OTHER "Geoengineering in Crisis: The Princeton Workshop on Geoengineering Ethics and Governance" 2024, Princeton, NJ

CONFERENCES & WORKSHOPS "Dynamics of the Global Water Cycle," 2022 Advanced Climate Dynamics Course (ACDC), Rondvassbu, Norway

2021 United Nations Climate Change Conference (**COP26**) Glasgow, Scotland

Inclusive Leadership Learning Cohort, Princeton GradFUTURES Fall 2020

COMPETENCIES **Languages** English (native), Chinese (intermediate), American Sign Language (advanced)

Computer Python, Matlab, R, $\text{\LaTeX}$, Microsoft Office

SELECTED **Princeton AOS Outreach** 2022–Present

SERVICE

Head Organizer, Volunteer

- Bronx Community College STEM Day (2025)
- Spring into Science at Princeton University (2025)
- Bronx Community College / City University of New York research scholars career presentation on climate & health (2024)
- NJ Ocean Fun Days (2024)
- Mercer County Boys & Girls Club Women and Girls Conference career panelist (2023), mentor (2024)
- Mercer County Boys & Girls Club Annual STEM Conference (2022, 2023)
- Bronx Community College climate science & policy career presentation (2022)

American Geophysical Union GeoHealth Early Career Committee 2025–Present
Committee Member

Princeton Women in Geosciences (PWIGS) 2022–Present
Mentor

Mentored Princeton University graduate students in Geosciences.

AOS Applicant Mentorship Program (AMP) 2022–2024

Organizer / Mentor

Helped prospective PhD students navigate the application process.

Association to Benefit Children – Study Buddies Connect

June 2021–Dec 2024

Volunteer Tutor / Mentor

Provided virtual and in-person one-on-one tutoring for a student for 2 hours/week in math, literacy, and science.

Princeton AOS Summer Workshop

2021, 2023

Organizer

- 2023: Paleo, present, and future: Leveraging the past to understand and predict our changing climate
- 2021: Climate Tipping Points

Princeton Undergraduates in Geosciences Mentorship Program

2020–2022

Graduate Mentor

Mentored Princeton University undergraduates in Geosciences and related fields.

Science Olympiad (Virginia, Massachusetts)

2017–2019

Event Supervisor, Test Writer

Ran Earth Science events for middle and high school students at regional and state competitions.

OTHER EXTRA-
CURRICULAR
ACTIVITIES

Princeton Energy and Climate Scholars

2023–2025

High Meadows Environmental Institute

- Helped run a three-part film screening and discussion series on the impacts of lithium mining on indigenous communities.
- Created educational plaques and helped establish a community microforest, along with mentoring volunteer students from Princeton Middle School.

Environmental Policy Associates Program

2023–Present

Center for Policy Research on Energy and the Environment